

3S25:
Introduction to
Fourier Analysis

Sam Condon

Impulse
Response and
Delta Functions

The Convolution
Integral

SAGI Summer School 2025: Introduction to Fourier Analysis

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- ① We have seen that the **transfer function** $H(\omega)$ is incredibly useful, describing the response of an LTI system to a complex exponential $e^{i\omega t}$ of arbitrary frequency ω .
- ② So far, we have been talking about the transfer function entirely in the frequency domain. Let's now investigate the time-domain representation of the transfer function.
- ③ To begin, let's define a function $h(t)$ as the **inverse fourier transform** of the transfer function $H(\omega)$:

$$h(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} H(\omega) e^{i\omega t} d\omega \quad (1)$$

That is, $h(t)$ is a summation of complex exponentials scaled at each frequency ω by the value of the transfer function at that frequency $H(\omega)$.

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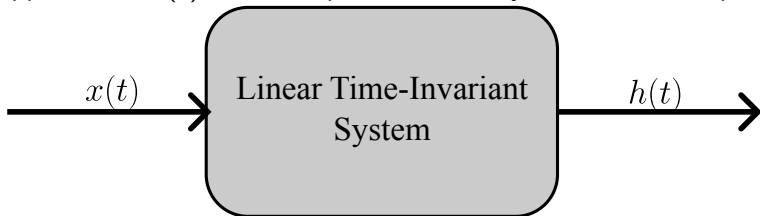
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- ❶ Now, suppose that $h(t)$ is the output of an LTI system to some input $x(t)$:



- ❷ Based on the form of 1 we can see that the inverse Fourier Transform expression for $x(t)$ is given by:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega t} d\omega \quad (2)$$

- ❸ This is equivalent to stating that the Fourier Transform of $x(t)$ is equal to 1 for all frequencies:

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt = 1 \quad \text{for all } \omega \quad (3)$$

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- ① Let's evaluate the integral of equation 2 to find out the explicit time-domain form of the input $x(t)$ which corresponds to the output $h(t)$. We start by splitting the integral into two components:

$$x(t) = \frac{1}{2\pi} \left[\int_{-\infty}^0 e^{i\omega t} d\omega + \int_0^{\infty} e^{i\omega t} d\omega \right] \quad (4)$$

$$= \int_0^{\infty} (e^{i\omega t} + e^{-i\omega t}) d\omega \quad (5)$$

- ② The indefinite integral of a complex exponential is quite straightforward. However, because of the upper bound of ∞ , we need to play some tricks to explicitly evaluate the above definite integral. Specifically, $e^{i\omega t} \big|_{\omega=\infty}$ is not a defined quantity.

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- ① The trick we play is to introduce an extra factor $e^{-\omega\epsilon}$ and take the limit $\epsilon \rightarrow 0^+$:

$$x(t) = \lim_{\epsilon \rightarrow 0^+} \int_0^{\infty} (e^{i\omega t} + e^{-i\omega t}) e^{-\omega\epsilon} d\omega \quad (6)$$

$$= \lim_{\epsilon \rightarrow 0^+} \int_0^{\infty} [e^{i\omega(t+i\epsilon)} + e^{-i\omega(t-i\epsilon)}] d\omega \quad (7)$$

$$= \lim_{\epsilon \rightarrow 0^+} \left[\frac{e^{i\omega t} e^{-\omega\epsilon}}{i(t+i\epsilon)} - \frac{e^{-i\omega t} e^{-\omega\epsilon}}{i(t-i\epsilon)} \right] \bigg|_0^{\infty} \quad (8)$$

- ② Now, because we have:

$$\lim_{\epsilon \rightarrow 0^+} e^{-\omega\epsilon} \bigg|_{\omega=\infty} = 0 \quad (9)$$

we obtain:

$$x(t) = \lim_{\epsilon \rightarrow 0^+} \left[\frac{i}{(t+i\epsilon)} - \frac{i}{t-i\epsilon} \right] = \lim_{\epsilon \rightarrow 0^+} \frac{2\epsilon}{t^2 + \epsilon^2} \quad (10)$$

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- ❶ Let's consider this result in two cases. The first being $t = 0$. Here, we have:

$$x(t = 0) = \lim_{\epsilon \rightarrow 0^+} \frac{2\epsilon}{\epsilon^2} = \lim_{\epsilon \rightarrow 0^+} \frac{2}{\epsilon} = \infty \quad (11)$$

- ❷ In the case $t \neq 0$, we have:

$$x(t \neq 0) = \lim_{\epsilon \rightarrow 0^+} \frac{2\epsilon}{t^2 + \epsilon^2} = \frac{0}{t^2 + 0} = 0 \quad (12)$$

- ❸ Summarizing this result:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega t} = \begin{cases} \infty, & t = 0 \\ 0, & t \neq 0 \end{cases} \quad (13)$$

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- ① There is one more interesting property of $x(t)$, and this is it's integral over all time:

$$\int_{-\infty}^{\infty} x(t) dt = ? \quad (14)$$

- ② Substituting equation 10:

$$\int_{-\infty}^{\infty} x(t) dt = \lim_{\epsilon \rightarrow 0^+} \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2\epsilon}{t^2 + \epsilon^2} dt \quad (15)$$

$$= \frac{1}{\pi} \tan^{-1} \left(\frac{t}{\epsilon} \right) \Big|_{-\infty}^{\infty} \quad (16)$$

$$= 1 \quad (17)$$

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- 1 In summary, we have the following result for an LTI system:

Result: Impulse Response

If the output of an LTI system $h(t)$ to some input $x(t)$ is such that $h(t)$ corresponds to the inverse Fourier Transform of the system transfer function $H(\omega)$, then the input $x(t)$ that produced the output $h(t)$ always obeys the following properties:

$$x(t) = \begin{cases} \infty, & t = 0 \\ 0, & t \neq 0 \end{cases} \quad (18)$$

$$\int_{-\infty}^{\infty} x(t) dt = 1 \quad (19)$$

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- ① Using the above properties of the input $x(t)$, we can prove a third property, which will uniquely define what is known as a delta function.
- ② Let's consider integrating the function $x(t)$, obeying 18 and 19, with an arbitrary non-zero function $f(t)$:

$$\int_{-\infty}^{\infty} f(t)x(t)dt = ? \quad (20)$$

- ③ Because $x(t)$ is zero everywhere, except for at $t = 0$, we obtain:

$$\int_{-\infty}^{\infty} f(t)x(t)dt = f(0) \int_{-\infty}^{\infty} x(t)dt = f(0) \quad (21)$$

where the final equality uses the integral property 19.

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- ① Repeating the previous integration, but with a time-shift in $x(t)$:

$$\int_{-\infty}^{\infty} f(t)x(t-T)dt = \int_{-\infty}^{\infty} f(\tau+T)x(\tau)d\tau = f(T) \int_{-\infty}^{\infty} x(\tau)d\tau = f(T) \quad (22)$$

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- ① Equations 18, 19, and 22 uniquely define what is known as a **Dirac Delta Function**. Repeating a full definition here:

Definition: Dirac Delta Function

A Dirac Delta Function $\delta(t)$ is a continuous function which obeys the following three properties:

$$\delta(t) = \begin{cases} \infty, & t = 0 \\ 0, & t \neq 0 \end{cases} \quad (23)$$

$$\int_{-\infty}^{\infty} \delta(t) dt = 1 \quad (24)$$

$$\int_{-\infty}^{\infty} f(t) \delta(t - T) dt = f(T) \quad (25)$$

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- 1 Repeating the above result again in terms of delta functions:

Result: Impulse Response

If the output of an LTI system $h(t)$ to some input $x(t)$ is such that $h(t)$ corresponds to the inverse Fourier Transform of the system transfer function $H(\omega)$, then the input $x(t)$ that produced the output $h(t)$ always corresponds to the delta function $\delta(t)$. The output of an LTI system to the delta function is called the **Impulse Response** of the system.

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- ① Now, suppose that we input a signal $x(t)$, which is a summation of two δ functions, separated by a time T :

$$x(t) = a\delta(t) + b\delta(t - T) \quad (26)$$

- ② The output signal corresponding to this input is:

$$y(t) = ah(t) + bh(t - T) \quad (27)$$

- ③ Re-arranging equation 25 a bit, we see that any input signal $x(t)$ can be expressed as a summation of scaled and shifted delta functions:

$$x(t) = \int_{-\infty}^{\infty} x(t')\delta(t' - t)dt' \quad (28)$$

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- ① The output associated with the input defined in equation 28 is then, through linearity and time-invariance:

$$y(t) = \int_{-\infty}^{\infty} x(t')h(t' - t)dt' \quad (29)$$

- ② The form of this integral is known as **the convolution integral** and we see that the output of any LTI system is the input signal convolved with the **impulse response** function.